🏦 Bank Transaction System using SQL Trigger

📌 Project Report

Project Title: Bank Transaction Management System using SQL Triggers
Technology: MySQL
Category: Database Mini Project

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1. Introduction

The Bank Transaction System is designed to manage account transactions such as deposits and withdrawals.

This project demonstrates the use of SQL triggers to automatically update account balances whenever a transaction occurs.

Triggers help automate database operations and ensure data consistency without manual intervention.

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2. Objectives

- To implement SQL triggers in a real-world scenario
- To automate account balance updates
- To maintain accurate transaction records
- To practice relational database design

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3. Database Design

Accounts Table

Stores customer account information.

Column	Data Type	Description
account_id	INT	Unique account ID
account_name	VARCHAR	Customer name
balance	INT	Current account balance

Query

```sql
CREATE TABLE accounts (
    account_id INT PRIMARY KEY,
    account_name VARCHAR(50),
    balance INT
);
```

Transactions Table

Stores deposit and withdrawal records.

Column	Data Type	Description
transaction_id	INT	Transaction ID
account_id	INT	Linked account
transaction_type	VARCHAR	Deposit / Withdraw
amount	INT	Transaction amount
transaction_date	DATETIME	Transaction time

Query

```sql
CREATE TABLE accounts (
    account_id INT PRIMARY KEY,
    account_name VARCHAR(50),
    balance INT
);
```

4. Data Insertion

Data Insertion Accounts Table

Query

```sql
INSERT INTO accounts VALUES
(1,'Rahim',50000),
(2,'Karim',30000),
(3,'Sakib',40000);
```

Query Result

account_id	account_name	balance
1	Rahim	50000
2	Karim	30000
3	Sakib	40000
NULL	NULL	NULL

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5. SQL Trigger Logic

A trigger is created to automatically update the account balance after a transaction is inserted.

Rules:

- If transaction type = **Deposit** → balance increases
- If transaction type = **Withdraw** → balance decreases

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6. Trigger Implementation

Query

```
DELIMITER $$

CREATE TRIGGER update_balance
AFTER INSERT ON transactions
FOR EACH ROW
BEGIN

IF NEW.transaction_type = 'Deposit' THEN
    UPDATE accounts
    SET balance = balance + NEW.amount
    WHERE account_id = NEW.account_id;

ELSEIF NEW.transaction_type = 'Withdraw' THEN
    UPDATE accounts
    SET balance = balance - NEW.amount
    WHERE account_id = NEW.account_id;

END IF;

END$$
```

7. Example Transaction

Result: Account balance automatically updates.

Deposit Example

```sql
INSERT INTO transactions
VALUES (1,1,'Deposit',10000,NOW());
```

account_id	account_name	balance
1	Rahim	60000
2	Karim	30000
3	Sakib	40000
NULL	NULL	NULL

Withdraw Example

```sql
INSERT INTO transactions VALUES
(2,1,'Withdraw',5000,NOW());
```

account_id	account_name	balance
1	Rahim	55000
2	Karim	30000
3	Sakib	40000
NULL	NULL	NULL

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8. Conclusion

This project demonstrates how SQL triggers can automate critical banking operations.
By automatically updating balances after transactions, the system maintains data accuracy and improves database efficiency.